Primordial Algorithm — Systems Dictionary and Missing Concepts Lexicon

0 . Purpose

This document acts as a **system dictionary** for the Primordial Algorithm project.

Its purpose is to collect, normalize, and clarify the concepts that were not fully formalized in architecture and source-spec documents.

This is **not** a new theory paper **not** and replacement for the architecture spec. It is **shared vocabulary layer** for:

- developers
- researchers
- contributors
- reviewers

The goal is to reduce ambiguity before implementation and community collaboration begin.

This document focuses especially on concepts that are often present implicitly in the model, but defined clearly enough for software construction.

---

1 . Why this document exists

The architecture spec defines:

- repository structure
- IPOD module boundaries
- system responsibilities
- anti-chaos engineering rules

The source spec defines:

- code skeletons
- algorithm entry points
- state objects
- execution flow

But neither document fully resolves all conceptual ambiguities.

A research simulation platform usually fails when:

- names are poetic but not operational
- one term is used in multiple senses
- mechanism and metaphor get mixed
- contributors implement different meanings under the same label

This document exists to prevent that drift.

---

2 . Dictionary Structure

Each entry contains some or all of the following fields:

- **Term**
- **Working definition**
- **Role in the system**
- **Software interpretation**
- **Scientific interpretation**
- **Common confusion**
- **Implementation note**

Not every entry needs every field, but this is the intended shape.

---

3 . Foundational System Terms

3 . 1 Primordial Algorithm

Working definition A research simulation framework for studying abiogenesis-like emergence using a dual-axis architecture that couples spatial field dynamics and temporal interaction readiness.

Role in the system This is the name of the whole platform, not the name of a single function algorithm.

Software interpretation A repository containing simulation, experiments, visualization, and analysis layers.

Scientific interpretation A model family rather than a single immutable law.

Common confusion Treating it as one equation or one universal theorem.

3.2 Model

Working definition A formalized but simplified representation of some class of processes.

Role in the system The project may contain multiple models over time:

- v 3 closed system
- v 4 open system
- future v 5 and beyond

Implementation note A model version must be traceable through presets, code version, and documentation.

3.3 Simulation

Working definition A computational execution of the model over time.

Role in the system Simulation is the runtime behavior produced by the model.

Common confusion Confusing the model itself with a single run of the model.

Implementation note Simulation state must be serializable.

3.4 World State

Working definition The full internal state required to continue a simulation from a given timestep.

Software interpretation Includes:

- field values
- entity states
- thermal state
- RNG state
- timestep
- sources
- metadata

Implementation note World state is the canonical source of truth for a run.

---

4. Ontology and Process Terms

4.1 V

Working definition The substrate layer from which activity can emerge and to which dissolution return.

Role in the system V functions as a foundational background of availability and inert potential.

Software interpretation V should not be treated as a mystical symbol. It may appear in at l distinct implementation senses:

1. **molecule type V** inside DNA or field composition
2. **field substrate concentration**
3. **thermodynamic background coupling**, if the model continues to link V conceptually to the bath

Common confusion Using the same variable name to mean both "one molecule class" and "the substrate ontology."

Implementation note Software should separate symbolic V from total substrate mechanics whenever possible.

4.2 Sinh

Working definition Emergence or appearance of a new active configuration.

Role in the system Represents threshold crossing into organized activity.

Software interpretation Sinh should not remain only a poetic term. In code it must correspond more measurable transitions, for example:

- first viable interaction
- first autocatalytic persistence event
- first stable network pattern

Common confusion Treating any state change as Sinh.

Implementation note Sinh should be operationalized through specific event classes or metrics.

4.3 Dẫn

Working definition Directed propagation driven by a combination of position-dependent opportunity ar time-accumulated readiness.

Role in the system It is the bridge between spatial logic and temporal logic.

Software interpretation Implemented through:

- gradient-following
- drive accumulation
- motion updates
- possibly future target selection or affinity weighting

Common confusion Reducing Dẫn to movement only.

Implementation note Dẫn is a process, not a one-step action.

4.4 Chuyển

Working definition The interaction trigger that fires when both spatial and temporal conditions satisfied.

Role in the system Acts as the gate between approach and transformation.

Software interpretation A conditional event boundary:

- overlap/proximity condition
- readiness/drive condition

Common confusion Treating Chuyển as generic “change.”

Implementation note Chuyển should be modeled as an event gate, not as a vague transformation bucket.

4.5 Dung

Working definition The integrative or structure-preserving component of interaction outcome.

Role in the system Represents consolidation, incorporation, or stabilization.

Software interpretation May affect:

- DNA extension

- energy gain
- complexity increase
- local field changes

Common confusion Assuming Dung means “good” or “successful.”

Implementation note Dung is a structural tendency, not a moral category.

4.6 Hoại

Working definition The dissolutive or structure-releasing component of interaction outcome.

Role in the system Represents erosion, release, turnover, or return to substrate.

Software interpretation May affect:

- DNA erosion
- field release
- breakdown events
- entropy-like turnover

Common confusion Assuming Hoại means “failure.”

Implementation note Hoại is not simply negative; it may be essential for adaptive turnover.

4.7 Dung/Hoại Ratio

Working definition The balance between integrative and dissolutive effects within one interaction.

Role in the system A central outcome descriptor.

Software interpretation A bounded pair of values that sum to 1.0.

Common confusion Treating the ratio as a fixed constant rather than a context-sensitive result.

---

5. Dual-Axis Terms

5.1 Spatial Axis

Working definition The layer that answers where conditions and entities are distributed.

Role in the system Provides gradients, proximity, dispersion, clustering, and neighborhood structure.

Software interpretation Implemented through:

- field coordinates
- concentration maps
- movement vectors
- distance checks

5.2 Temporal Axis

Working definition The layer that answers when interaction becomes possible or meaningful.

Role in the system Tracks accumulation, readiness, sequence, persistence, and delayed activation.

Software interpretation Implemented through:

- timesteps
- drive accumulation
- event order
- time-based thresholds

5.3 Dual-Axis Coupling

Working definition The principle that meaningful interaction requires both spatial and temporal conditions.

Role in the system This is one of the central design commitments of the model.

Implementation note If the software allows one axis to dominate silently, the model collapses weaker form.

5.4 Readiness

Working definition A time-built capacity or pressure that makes interaction possible.

Software interpretation Usually represented `drive_acc`, but the concept is broader than one variable.

Common confusion Assuming readiness is equivalent to energy.

Implementation note Readiness may depend on multiple state components in future versions.

5.5 Overlap

Working definition The condition of spatial co-presence within an interaction radius.

Software interpretation Usually a distance threshold.

Common confusion Treating overlap as sufficient by itself.

---

6. Thermodynamic and Energy Terms

6.1 Delta-G (`ΔG`)

Working definition A thermodynamic driver used to estimate whether an interaction is favorable unfavorable under current temperature conditions.

Role in the system Provides the physical-style grounding of Dung/Hoại balance.

Software interpretation A pure function.

Common confusion Over-reading the toy implementation as literal chemistry.

Implementation note This is a model-level thermodynamic abstraction, not a claim of direct bioche realism.

6.2 Temperature

Working definition A control variable that modulates interaction energetics and noise conditions.

Role in the system Affects spontaneity and possibly future disorder, stochasticity, and timescale behavior.

Software interpretation Should be encapsulat `ThermalBath`, not scattered as loose parameters everywhere.

6.3 Heat Bath

Working definition The thermodynamic background condition of the system.

Role in the system Provides the temporal environment in which interactions occur.

Common confusion Treating heat bath and molecule concentration substrate as the same thing in code.

6.4 Open System

Working definition A system receiving ongoing external energy input.

Role in the system Necessary for sustained non-equilibrium behavior.

Software interpretation Represented through energy source classes.

6.5 Closed System

Working definition A system without external replenishing input.

Role in the system Useful as a baseline or control comparison.

6.6 Energy Source

Working definition Any mechanism that injects available energetic potential into the simulation.

Software interpretation A plugin-like source type such as:

- hydrothermal vent
- UV source
- future lightning source

Implementation note Energy sources should mutate the world only through defined state interfaces.

6.7 Conservation Drift

Working definition The deviation between expected and observed energy-like bookkeeping over time.

Role in the system A diagnostic term, not a primary ontology term.

Implementation note Needed to distinguish model behavior from accounting failure.

---

7 . Entity and DNA Terms

7 . 1 Entity

Working definition A stateful agent or structured unit participating in the simulation.

Role in the system The main carrier of DNA, energy, motion, interaction, and emergence.

Software interpretation A mutable domain object, but mutation must remain controlled.

---

7 . 2 DNA

Working definition A symbolic sequence representing the internal structured composition of an entity.

Role in the system Acts as a compact representation of interaction tendencies and structural identity.

Common confusion Reading it as literal biological DNA.

Implementation note It is better understood as a symbolic internal composition vector or sequence.

---

7 . 3 DNA Resonance

Working definition A compatibility or complementarity score between symbolic sequences.

Role in the system Used in some versions to estimate relational affinity.

Implementation note Resonance should be treated as a model metric, not as literal physical res
unless later justified more rigorously.

---

7 . 4 DNA Stability

Working definition A score estimating structural integrity across an entity's internal sequence.

Role in the system Used in division and emergence eligibility.

Common confusion Confusing stability with fitness.

7.5 Mutation

Working definition A stochastic alteration in symbolic internal composition.

Role in the system Introduces variation.

Implementation note Mutation must always be tied to seeded randomness.

7.6 Level

Working definition A discrete state marker representing developmental stage or structural class.

Role in the system Allows the model to distinguish between forms such as molecule, network, vesic proto-cell.

Common confusion Assuming level equals objective biological category.

Implementation note Levels are model-internal categories and must be documented as such.

---

8. Emergence Terms

8.1 Emergence

Working definition The appearance of higher-order organized behavior or structure from lower-level l interactions.

Role in the system One of the main phenomena the simulation tries to study.

Common confusion Using “emergence” for anything visually interesting.

Implementation note Emergence claims should be tied to explicit metrics or event definitions.

8.2 Emergence Rule

Working definition A software criterion that upgrades an entity or state into a new class.

Role in the system An implementation convenience that may or may not correspond to “true emergence.”

Important distinction A threshold rule is not the same thing as an emergent mechanism.

8.3 True Emergence

Working definition A stronger claim that the system's dynamics themselves generate a novel organi without hand-authored promotion logic.

Implementation note This should not be claimed casually.

8.4 Complexity

Working definition A scalar measure intended to approximate structured richness or organization.

Role in the system Used in promotion and evaluation.

Common confusion Using the same complexity score as both cause and proof of emergence.

Implementation note Mechanism, transition rule, and evaluation metric should be kept separate.

8.5 Persistence

Working definition The ability of an entity or pattern to survive over time.

Role in the system A critical but under-formalized dimension of emergence.

Implementation note Persistence deserves its own metric instead of being folded vaguely into complexity.

8.6 Autocatalytic Pattern

Working definition A self-supporting interaction pattern that helps maintain or reproduce its own en conditions.

Role in the system This is one of the most important missing formal concepts for future development.

Implementation note Future versions should define explicit detection criteria for autocatalytic loops self-reinforcing motifs.

9 . Spatial Dynamics Terms

9 . 1 Gradient

Working definition A directional signal pointing toward increasing local concentration or opportunity.

Role in the system Guides movement and therefore shapes interaction probability.

---

9 . 2 Diffusion

Working definition The spreading of concentration over neighboring space.

Role in the system Creates background redistribution and prevents static concentration maps.

Common confusion Treating diffusion as just visual smoothing.

---

9 . 3 Clustering

Working definition The tendency of entities to become spatially concentrated.

Role in the system Can be either an interesting emergent behavior or an artifact.

Implementation note The system must explicitly distinguish:

- meaningful clustering
- pathological over-convergence

---

9 . 4 Spatial Exclusion

Working definition A rule preventing unlimited co-occupation of the same spatial region.

Role in the system A missing realism constraint in simpler model versions.

Implementation note Should likely enter as a repulsion term or occupancy constraint.

---

9 . 5 Competition for Space

Working definition The idea that local occupation itself constrains future occupation and access.

Role in the system Necessary for preventing collapse into unrealistic clustering.

9.6 Spatial Entropy

Working definition A measure of how dispersed or concentrated entities or field values are acro
environment.

Role in the system A useful analysis term missing from earlier specs.

---

10. Time and Scheduler Terms

10.1 Timestep

Working definition One discrete update cycle of the simulation.

Role in the system The base temporal unit for orchestration.

10.2 Tick Pipeline

Working definition The ordered sequence of operations executed per timestep.

Role in the system Critical for scientific validity.

Implementation note Tick order is part of the model, not just a coding convenience.

10.3 Scheduler

Working definition The component that controls execution order and future multi-timescale behavior.

Role in the system Protects model semantics from accidental reordering.

10.4 Timescale Hierarchy

Working definition The idea that different processes may operate at different update frequencies.

Role in the system Important future extension for improving realism.

---

11. Recursion and Fractality Terms

11.1 Fractal Cycle

Working definition A recursive evaluation process in which the core interaction logic is re-applied a
sub-levels.

Role in the system Encodes the self-similarity claim of the framework.

Common confusion Equating “recursive” with “physically proven fractal.”

Implementation note The code can be recursive even if the scientific meaning remains provisional.

11.2 Depth

Working definition The current level of recursive evaluation.

Role in the system Controls cost, attenuation, and temperature scaling.

11.3 Attenuation

Working definition The weakening of contribution from deeper recursive calls.

Role in the system Keeps recursion from dominating total outcome unrealistically.

11.4 Recursive Cost

Working definition The computational overhead introduced by recursive branching.

Role in the system A software engineering term as much as a modeling term.

Implementation note Benchmarking recursion is mandatory, not optional.

---

12. Analysis and Evaluation Terms

12.1 Metric

Working definition A measurable quantity used to describe some aspect of simulation behavior.

Role in the system Metrics support analysis, comparison, and claims.

Common confusion Assuming metrics are neutral.

Implementation note Metrics shape what the team sees, so they influence research direction.

12.2 Diagnostic

Working definition A measurement or rule used primarily to detect artifacts, bugs, or suspicio behaviors.

Role in the system Different from scientific metrics aimed at explaining the phenomenon.

12.3 Artifact

Working definition A behavior produced by the implementation or simplification rather than the int modeled principle.

Role in the system Artifact detection is central to trustworthy simulation research.

12.4 Baseline

Working definition A reference model or run used for comparison.

Role in the system Prevents arbitrary interpretation of new outcomes.

12.5 Ablation

Working definition A controlled removal or weakening of one model component to see what changes.

Role in the system One of the most important methods for understanding causal importance.

12.6 Benchmark

Working definition A standardized run or measurement used to compare versions, settings, or implementations.

Role in the system Supports reproducibility and engineering discipline.

---

13. Software Architecture Terms

13.1 Input

Working definition The layer responsible for ingestion, validation, and normalization of incoming configuration or control signals.

Role in the system The entry boundary to the simulation platform.

13.2 Process

Working definition The layer containing core business logic or model logic.

Role in the system The most protected layer in the architecture.

13.3 Output

Working definition The layer responsible for externalization:

- logs
- files
- exports
- checkpoints

Common confusion Letting output code mutate scientific state.

13.4 Display

Working definition The rendering and interface layer.

Role in the system Shows what happened, but does not define what happened.

13.5 Data

Working definition The analysis and learning layer.

Role in the system Transforms raw run history into interpretable information.

13.6 Single Source of Truth

Working definition The principle that one canonical representation of simulation state must be authoritative.

Role in the system Prevents software-level chaos.

13.7 Determinism Boundary

Working definition The line separating what must remain reproducible from what may vary stochas under seeded control.

Role in the system A missing but essential term for research software.

13.8 Legitimate Simulation Chaos

Working definition Model-allowed stochasticity that belongs to the phenomenon being studied.

Examples - mutation - random emission - diffusion variation - sensitivity to initial conditions

---

13.9 Illegitimate Software Chaos

Working definition Uncontrolled disorder introduced by bad software design.

Examples - hidden randomness - race conditions - UI state drift - callback-dependent results - world states

---

13.10 Epistemic Corruption

Working definition A state in which the software no longer allows us to tell whether a result the model or from implementation noise.

Role in the system This is one of the most important cautionary terms for the entire project.

---

14. Missing Concepts That Should Be Formalized Later

This section lists the most important concepts that still need stronger treatment in future versions.

14.1 Fitness Landscape

A more rigorous way to describe why certain internal compositions survive more often than others.

14.2 Resource Scarcity

A more explicit model of local depletion and competition, beyond generic field concentration.

14.3 Membrane Boundary

A real distinction between inside and outside, rather than just a higher level state label.

14.4 Internal vs External Chemistry

An explicit separation of reactions happening within an entity and reactions happening in ambient space.

14.5 Persistence Metric

A direct measurement of how long structures maintain identity.

14.6 Autocatalytic Loop Detection

A concrete method for identifying self-reinforcing reaction or interaction motifs.

14.7 Information-Theoretic Measures

Metrics such as mutual information, predictive alignment, or environmental coupling.

14.8 Occupancy Constraint

A formal rule for limited local space capacity.

14.9 Adaptive Thresholding

The possibility that some thresholds should emerge from state rather than remain fixed constants.

14.10 Identity Drift

A way to measure how much an entity remains “the same” across many interactions.

---

15. Terms That Should Be Used Carefully

15.1 Life

Use carefully. The current system studies life-like or abiogenesis-like processes, not life in the biological sense.

15.2 Consciousness

Use very carefully. Earlier versions use consciousness language, but unless the metric is well-defin term should remain provisional.

15.3 Universal

Avoid using this casually. A model may be broad, suggestive, or powerful without being universal.

15.4 Objective

The system may simulate objective-style constraints, but outputs remain model-dependent.

15.5 Physical

The model can be physics-inspired without being literally physically validated.

---

16. Naming Policy Recommendations

To avoid confusion in the codebase:

- use **conceptual names** in docs
- use **operational names** in code
- keep an explicit mapping table

Example:

Concept term	Operational code term
Dẫn	`drive_accumulation` / `dan()`
Chuyển	`interaction_gate` / `chuyen()`
Dung	`integration_ratio`
Hoại	`dissolution_ratio`
V	`substrate_symbol` or `field_V` depending on usage

This mapping should become part of contributor onboarding.

---

17. Final Interpretation

This dictionary is a stabilizing layer.

It does not prove the theory. It does not replace the architecture. It does not replace code.

What it does is much more practical:

- it reduces ambiguity
- it clarifies terms before coding
- it protects collaboration from vocabulary drift
- it helps separate metaphor from mechanism
- it makes future extension more disciplined

For a project like this, a shared dictionary is not optional. It is part of the architecture.

---

18. Immediate Next Step

After this document, one useful follow-up could be a **Concept-to-Code Mapping Table** that lists, for each major term:

- theory label
- code module
- data fields
- metrics affected
- tests required

That would make the transition from vocabulary to implementation even tighter.